

enable your billing. The platform provides a trial period of two months for US\$300, which is sufficient to try out the APIs.

Once you have signed up for the platform, you need to create a project. Go to the *Cloud Console* and click on *New Project*. In the *New Project* dialogue box, provide a name for your project, as shown in Figure 1, before clicking on *Create*.

Once the project is created, you need to enable the above three APIs for your project. To do that, go to the *Main Menu* on the top left in the *Cloud Console* and click on *API Manager*, as shown in Figure 2.

In the search bar, enter 'Cloud Vision' and it will display the Google Cloud Vision API in the list. Select it and then click on *Enable*. This will enable the API. Perform the same operation for the other two APIs also, i.e., the natural language processing (NLP) API and the Speech API. Ensure that you have enabled the above three APIs for your project before moving ahead.

Since we are going to be writing an external program that invokes the REST APIs, our calling client application needs to identify itself to the Google Cloud platform. We do that through the concept of a service account, which will be used by our client application to invoke the APIs.

To do that, we need to download the service account keys for our project. From the *API Manager* menu on the left, click on *Credentials* and then on *Create Credentials*. Select *Service Account Key* from the options. This will bring up a dialogue box as shown in Figure 3. Select *New Service*

Figure 1: *New Project* dialogue box

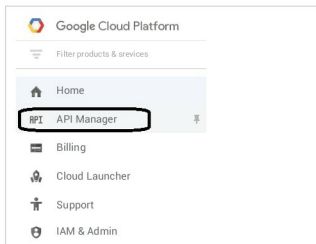


Figure 2: *API Manager*

Account, give a name to the service account and in *Role*, select *Project Owner*.

Click on *Create*. This will download a JSON file for your service account. Ensure that you save this file in a safe place, since we will use it when we run our client application. For the purpose of this article, the service account JSON file downloaded will be referred to as the *mlapi-serviceaccount.json* file.

Vision API

Let's first test the vision API. As mentioned earlier, this has multiple features and, in this specific example, we are going to test its label detection feature. This feature allows you to provide an image, and it will identify various labels for this image, which can help in its classification.

We are going to provide it an image of a cricket match and see what it can figure out about it. The image that we are going to test it against is shown in Figure 4.

Before you run this code and the other examples, you will need to set an environment variable named `GOOGLE_APPLICATION_CREDENTIALS` and it will need to point to the service account

Figure 3: *Service Account Key* creation

JSON file that we downloaded earlier. For example:

```
export GOOGLE_APPLICATION_CREDENTIALS=mlapi-
serviceaccount.json
```

The sample code in Python is shown below. This is a modification of the code that is available at its API page.

```
from googleapiclient import discovery
import argparse
import base64
import json
from oauth2client.client import GoogleCredentials
```